

Name: Maths Teacher: *File*

SYDNEY TECHNICAL HIGH SCHOOL



Year 11

Mathematics

Assessment 2

JULY, 2015

Time allowed: 90 minutes

○ ***General Instructions:***

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided

- Section 1 Multiple Choice
 - Questions 1-5
 - 5 Marks
- Section II Questions 6-13
 - 63 Marks

Section I

Answers to be done on the multiple choice answer sheet in your answer booklet.

1. What are the solutions of $2x^2 - 5x - 1 = 0$?

(A) $x = \frac{-5 \pm \sqrt{17}}{4}$

(B) $x = \frac{5 \pm \sqrt{17}}{4}$

(C) $x = \frac{-5 \pm \sqrt{33}}{4}$

(D) $x = \frac{5 \pm \sqrt{33}}{4}$

2. Which inequality defines the domain of the function $f(x) = \frac{1}{\sqrt{x+3}}$?

(A) $x > -3$

(B) $x \geq -3$

(C) $x < -3$

(D) $x \leq -3$

3. Find the values of m for which $24 + 2m - m^2 \leq 0$

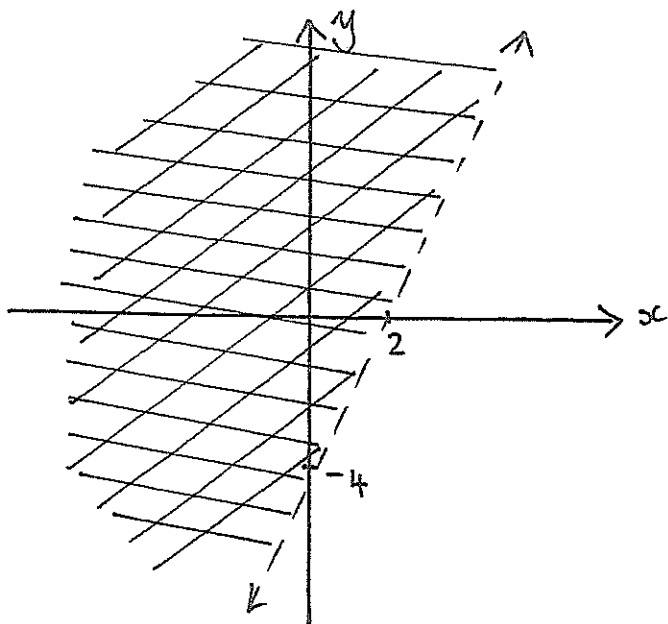
(A) $m \leq -4$ or $m \geq 6$

(B) $m \leq -6$ or $m \geq 4$

(C) $-4 \leq m \leq 6$

(D) $-6 \leq m \leq 4$

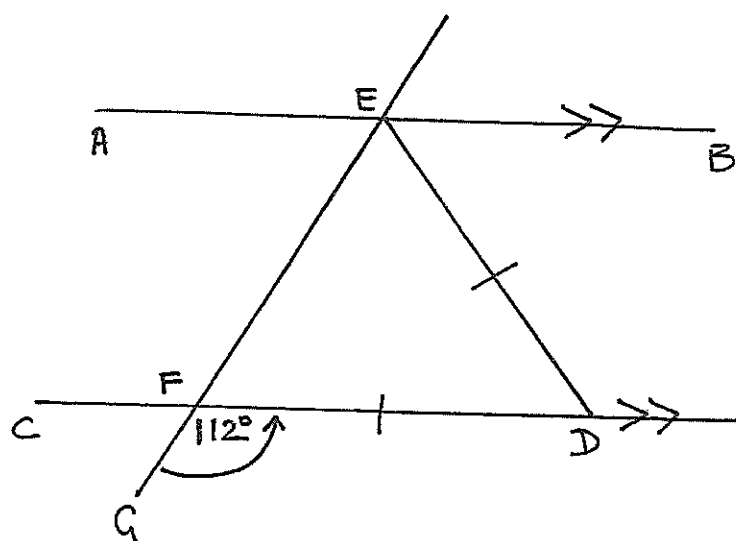
4.



The shaded region is best described by the inequality.

- (A) $2x - y - 4 \geq 0$
- (B) $2x - y - 4 \leq 0$
- (C) $2x - y - 4 > 0$
- (D) $2x - y - 4 < 0$

5.



If $AB \parallel CD$, $ED = FD$ and $\angle DFG = 112^\circ$
then $\angle BED =$

- (A) 112°
- (B) 24°
- (C) 68°
- (D) 44°

Section II

Mark

Question 6 – (8 marks)

- a) Evaluate $\sqrt[3]{\frac{651}{4\pi}}$ to four significant figures
- b) Solve $2 - 3x \leq 8$ and sketch your solution on a number line
- c) Solve $x^2 - 6x = 0$
- d) Solve $4 < 4x - 3 < 9$

2

2

2

2

Question 7 – (8 marks) – Start a new page

- a) Express $\frac{a^{-1}+b^{-1}}{a+b}$ in simplest fraction form without using negative indices.
- b) Solve $|5x - 2| = |3x + 4|$
- c) Solve $\frac{5}{8}(x+4) = 4x - \frac{1}{2}$
- d) Express $\frac{3\sqrt{2}}{3\sqrt{2}+2\sqrt{3}}$ in the form $a + b\sqrt{6}$

2

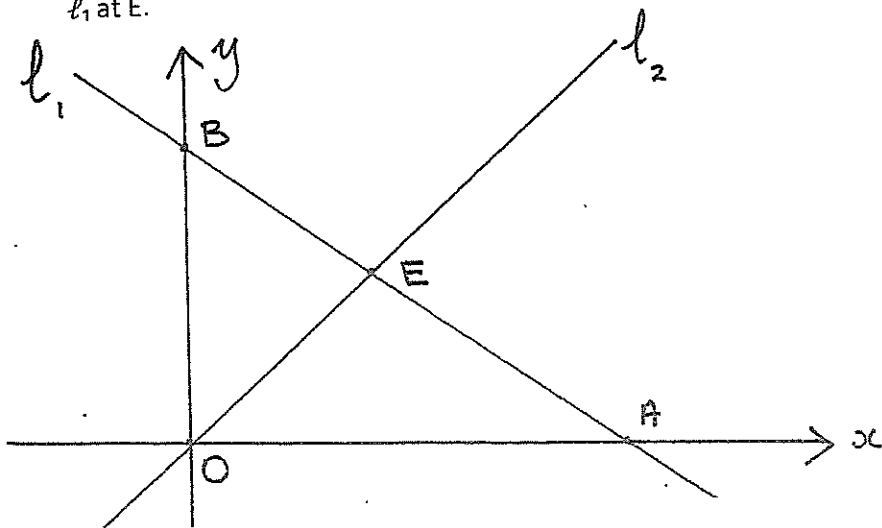
2

2

2

Question 8 – (8 marks) – Start a new page

- a) The diagram shows a line ℓ_1 with equation $3x + 4y - 12 = 0$, which intersects the y axis at B.
A second line ℓ_2 with equation $4x - 3y = 0$, passes through the origin O and intersects ℓ_1 at E.



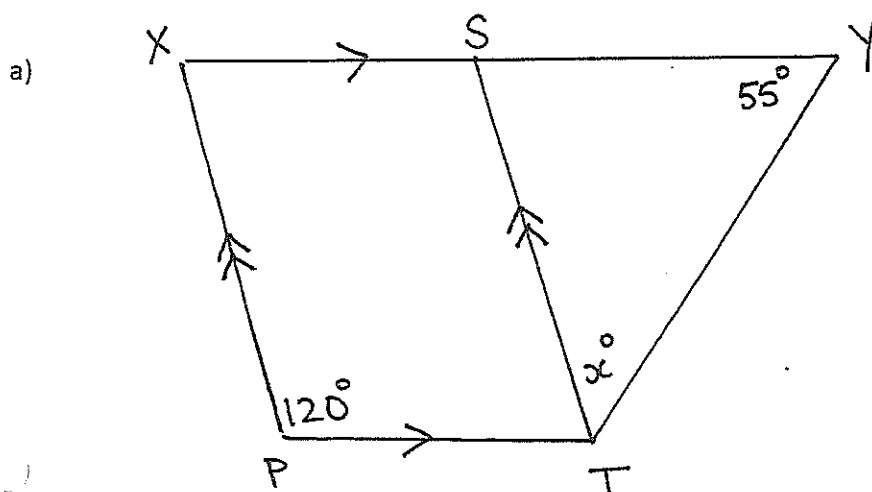
NOT TO SCALE

- (i) Show that coordinates of B are (0, 3). 1
- (ii) Show that ℓ_1 is perpendicular to ℓ_2 . 2
- (iii) Show that the perpendicular distance from O to ℓ_1 is $\frac{12}{5}$ units. 1
- (iv) Using Pythagoras' theorem, or otherwise, find the length of the interval BE. 1
- (v) Hence, or otherwise, find the area of $\triangle BOE$. 1

b) Simplify $\frac{x^3-1}{x^2-1} \div \frac{3x^2+3x+3}{x^2-4x-5}$ 2

Question 9 – (7 marks) – Start a new page

Mark



$XY \parallel PT$ and $XP \parallel ST$

Redraw the diagram in your answer booklet.

Find x giving reasons for your answer.

- b) A function is defined as follows

$$f(x) = \begin{cases} 0 & \text{if } x \leq -3 \\ -1 & \text{if } -3 < x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Find

i) $f(-3) + f(-2) + f(2)$ 1

ii) $f(a^2)$ 1

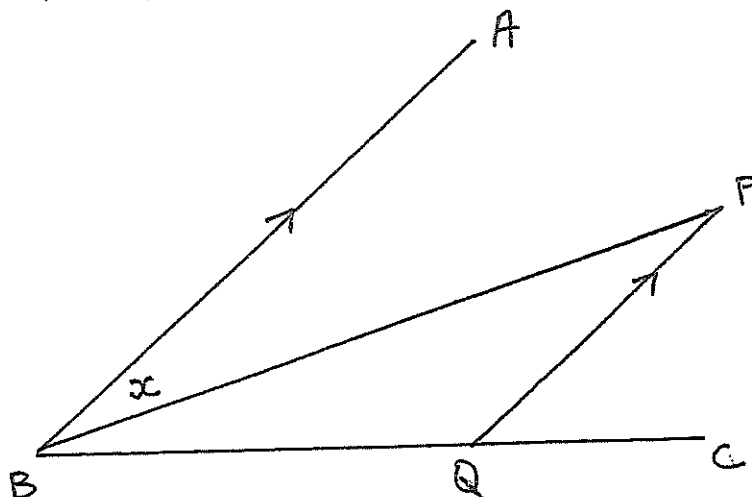
- c) i) Sketch $y = |x - 1|$ and $y = x + 1$ on the same axes.
Use a ruler and label each function carefully. Show any points of intersection with the x and y axes. Your sketch should be approximately half a page.
- ii) Hence solve $|x - 1| > x + 1$

2

1

Question 10 – (8 marks) – Start a new page

a)



2

Let $\angle ABP = x$

BP bisects $\angle ABC$ and $AB \parallel PQ$

Redraw this diagram in your answer booklet. Use a ruler.

Your diagram should be approximately half a page in size.

Prove that $BQ = PQ$

- b) Find the exact value of

1

i) $\sin 225^\circ$

ii) $\tan(-30^\circ)$

1

- c) If θ is obtuse and $\tan \theta = -\frac{1}{5}$ find the exact value of $\cos \theta$

1

- d) Prove $\frac{1}{\sin \theta \cos \theta} - \tan \theta = \cot \theta$

3

Question 11 – (8 marks) – Start a new page

- a) Solve the following in the domain $0^\circ \leq x \leq 360^\circ$.
(write your answers correct to the nearest minute)

i) $\tan 2\theta = -1$ 2

ii) $3 \sin^2 \theta + 2 \sin \theta = 0$ 2

iii) $3 \sin \theta = 2 \cos \theta$ 2

b) Find $\lim_{x \rightarrow 3} \frac{x-3}{x^2-9}$ 2

Question 12 – (8 marks) – Start a new page

- a) Differentiate the following

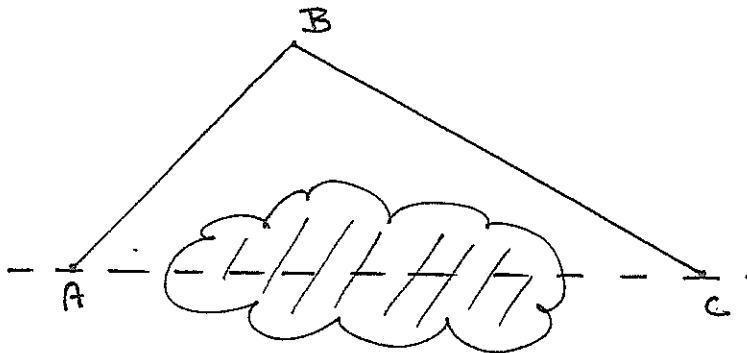
i) $y = 4x^3 - x + 5$ 1

ii) $y = (3x^2 - 4)^4$ 2

iii) $y = \frac{x+1}{x-1}$ 2

- b)

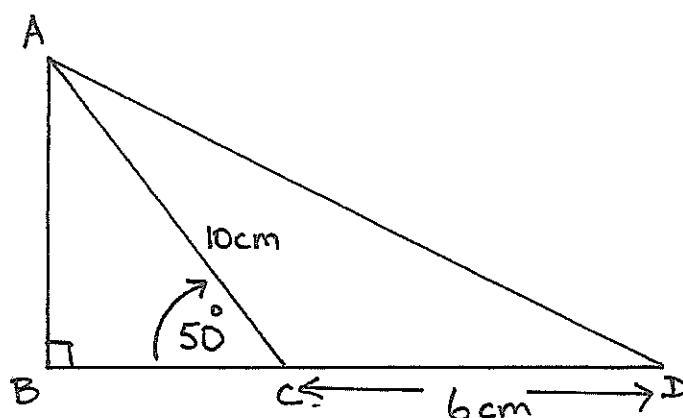
A surveyor walking due east turns at A to avoid marshy country and walks 270 metres to B on a bearing of 048° and then turns and walks on a bearing of 112° to C. C is due east of A.



- i) Redraw the diagram showing the size of angles $\hat{BAC}$, $\hat{ABC}$ and $\hat{BCA}$. 1

- ii) Hence find the length of AC to the nearest metre. 2

- a) In the figure $CD = 6\text{cm}$, $AC = 10\text{cm}$, angle $ACB = 50^\circ$ and angle $ABC = 90^\circ$. Find:



- i) AD to the nearest cm

2

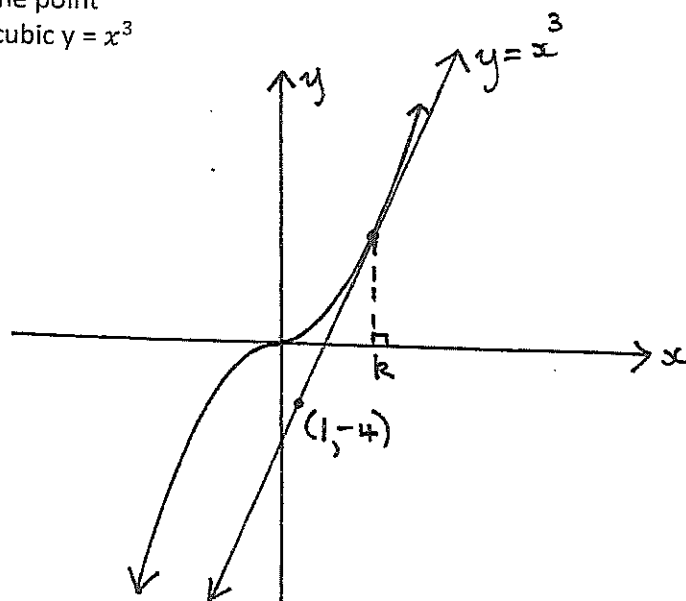
- ii) Area of $\triangle ACD$ to the nearest cm^2 .

1

- b) i) Show that $k = 2$ is a solution to the equation $2k^3 - 3k^2 - 4 = 0$

1

- ii) The diagram shows a tangent at the point where $x = k$ (where $k > 0$) to the cubic $y = x^3$



- α . Find the gradient of the tangent at $x = k$

1

- β . Find the equation of the tangent at $x = k$

2

- γ . If the tangent is found to pass through $(1, -4)$ find the value of k .

1

Q1	D
2	A
3	A
4	D
5	D

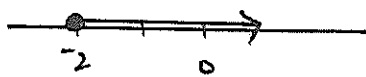
Question 6

a) $\underline{3.728}$ (4 sig. fig)

b) $2 - 3x \leq 8$

$-3x \leq 6$

$x \geq -2$



c) $x^2 - 6x = 0$

$x(x - 6) = 0$

$\therefore x = 0, x = 6$

d) $4 < 4x - 3 < 9$

$7 < 4x < 12$

$\underline{\underline{\frac{7}{4} < x < 3}}$

Question 7

a) $\left(\frac{1}{a} + \frac{1}{b}\right) \div (a+b)$

$\frac{(b+a)}{ab} \times \frac{1}{(a+b)}$

$\underline{\underline{\frac{1}{ab}}}$

b) $5x - 2 = 3x + 4$ $5x - 2 = -(3x + 4)$

$2x = 6$

$5x - 2 = -3x - 4$

$\therefore x = 3$

$8x = -2$

and

$\underline{\underline{x = -\frac{1}{4}}}$

c) $\frac{5}{8}(x+4) = 4x - \frac{1}{2}$

$5(x+4) = 32x - 4$

$5x + 20 = 32x - 4$

$24 = 27x$

$x = \frac{24}{27}$

$\therefore x = \underline{\underline{\frac{8}{9}}}$

d) $\frac{3\sqrt{2}}{3\sqrt{2}+2\sqrt{3}} \times \frac{3\sqrt{2}-2\sqrt{3}}{3\sqrt{2}-2\sqrt{3}}$

$\frac{3\sqrt{2}(3\sqrt{2}-2\sqrt{3})}{18-12}$

$18-12$

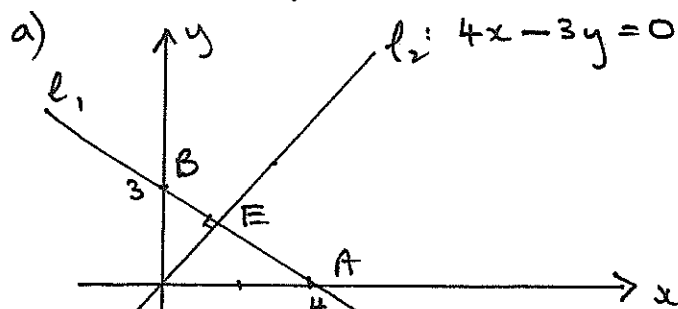
$18-6\sqrt{6}$

6

$\underline{\underline{\frac{6(3-\sqrt{6})}{6}}}$

$\therefore \frac{3\sqrt{2}}{3\sqrt{2}+2\sqrt{3}} = 3 - \sqrt{6}$

Question 8



i) sub. $x=0$ into $l_1: 3x+4y-12=0$
 $4y = 12$
 $y = 3$

$\therefore B(0, 3)$

ii) $m_{l_1} = -\frac{3}{4}$ $m_{l_2} = \frac{4}{3}$

since $-\frac{3}{4} \cdot \frac{4}{3} = -1$

$\therefore \underline{\underline{l_1 \perp l_2}}$

$$ii) P = \left| \frac{3 \cdot 0 + 4 \cdot 0 - 12}{\sqrt{9+16}} \right|$$

$$l_1: 3x + 4y - 12 = 0$$

$$P = \left| \frac{-12}{5} \right|$$

$$\therefore P = \frac{12}{5} \text{ units}$$

$$iv) \begin{array}{l} \text{Diagram: A right triangle BOE with right angle at E. Side BO = 3, side OE = 12/5.} \\ 3^2 - \left(\frac{12}{5}\right)^2 = (BE)^2 \\ 9 - \frac{144}{25} = (BE)^2 \\ \frac{81}{25} = (BE)^2 \end{array}$$

$$\therefore BE = \frac{9}{5} \text{ units}$$

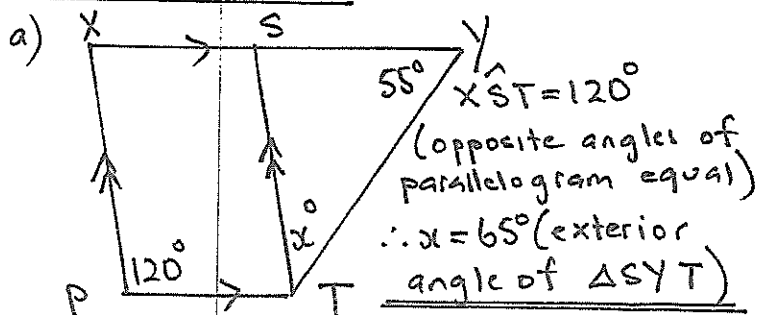
$$v) \text{ Area } \triangle BOE = \frac{1}{2} \left(\frac{12}{5} \times \frac{9}{5} \right) = \frac{54}{25} \text{ units}^2$$

$$b) \frac{x^3 - 1}{x^2 - 1} \times \frac{x^2 - 4x - 5}{3x^2 + 3x + 3}$$

$$\frac{(x-1)(x^2+x+1) \times (x-5)(x+1)}{(x-1)(x+1) \cdot 3(x^2+x+1)}$$

$$\frac{x-5}{3}$$

Question 9

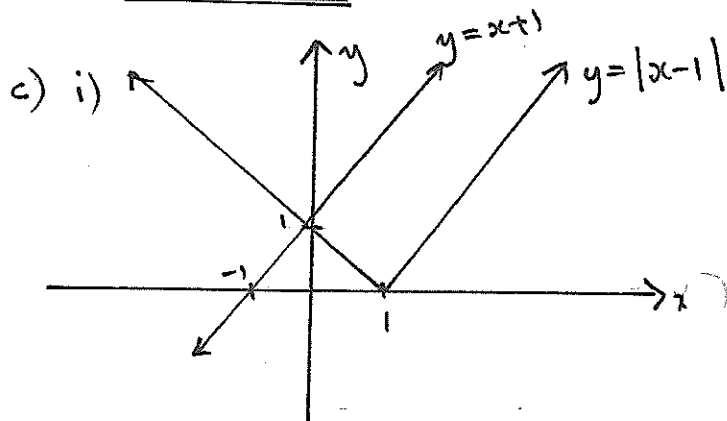


$$b) i) f(-3) + f(-2) + f(2)$$

$$= 0 + -1 + 2$$

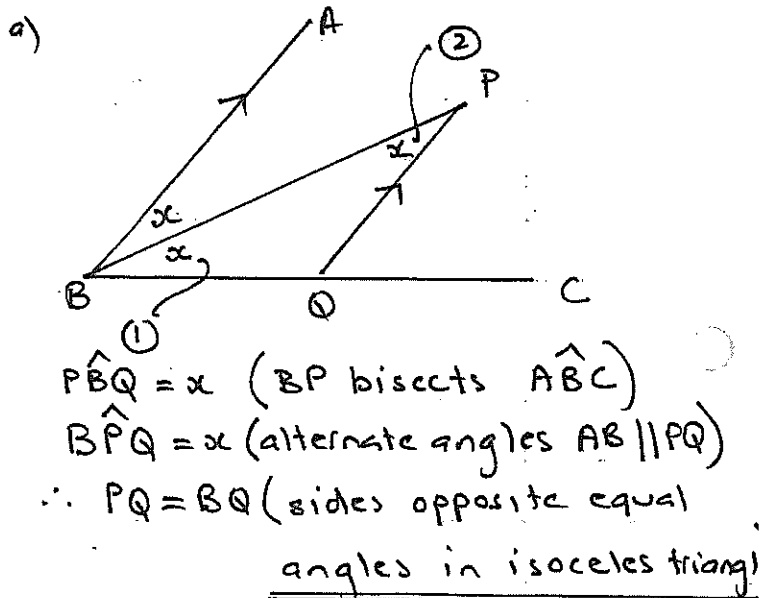
$$= 1$$

$$ii) \underline{f(a^2) = a^2} \text{ since } a^2 \geq 0$$



$$ii) \underline{x < 0}$$

Question 10



$$b) i) \sin 225^\circ = \sin (180 + 45)$$

$$\begin{array}{c|c} S & A \\ \hline T & C \end{array}$$

$$= -\sin 45^\circ$$

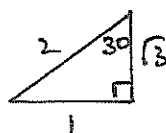
$$= -\frac{1}{\sqrt{2}}$$

$$ii) \tan(-30^\circ) = \tan(360 - 30)$$

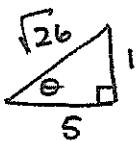
$$\begin{array}{c|c} S & A \\ \hline T & C \end{array}$$

$$= -\tan 30^\circ$$

$$= -\frac{1}{\sqrt{3}}$$



c) $\tan \theta = -\frac{1}{5}$ $\begin{array}{c|c} \checkmark S & A \\ \hline T & C \checkmark \end{array}$



$\therefore \cos \theta = -\frac{5}{\sqrt{26}}$

d) LHS = $\frac{1}{\sin \theta \cdot \cos \theta} - \tan \theta$

$$= \frac{1}{\sin \theta \cdot \cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\sin \theta \cdot \cos \theta}$$

$$= \frac{\cancel{\cos \theta}}{\cancel{\cos \theta} \sin \theta}$$

$$= \frac{\cos \theta}{\sin \theta}$$

$$= \cot \theta$$

= RHS

QUESTION 11

a) i) $\tan 2\theta = -1$ $\begin{array}{c|c} \checkmark S & A \\ \hline T & C \checkmark \end{array}$

"acute" $2\theta = 45^\circ$

$\therefore 2\theta = 135^\circ, 315^\circ, 495^\circ, 675^\circ$

$\theta = 67\frac{1}{2}^\circ, 157\frac{1}{2}^\circ, 247\frac{1}{2}^\circ, 337\frac{1}{2}^\circ$

OR $67^\circ 30', 157^\circ 30', 247^\circ 30', 337^\circ 30'$

ii) $3 \sin^2 \theta + 2 \sin \theta = 0$

$$\sin \theta (3 \sin \theta + 2) = 0$$

$\sin \theta = 0$ $\sin \theta = -\frac{2}{3}$

$\theta = 0^\circ, 180^\circ, 360^\circ$ and $\begin{array}{c|c} S & A \\ \hline \checkmark T & C \checkmark \end{array}$

$\theta = 221^\circ 49', 318^\circ 11'$

iii) $3 \sin \theta = 2 \cos \theta$

$\frac{\sin \theta}{\cos \theta} = \frac{2}{3}$

$\tan \theta = \frac{2}{3}$ $\begin{array}{c|c} S & A \\ \hline \checkmark T & C \checkmark \end{array}$

$\therefore \theta = 33^\circ 41', 213^\circ 41'$

b) $\lim_{x \rightarrow 3} \frac{1(x-3)}{(x-3)(x+3)}$

$$= \frac{1}{6}$$

Question 12.

a) i) $\frac{d}{dx} (4x^3 - x + 5) = 12x^2 - 1$

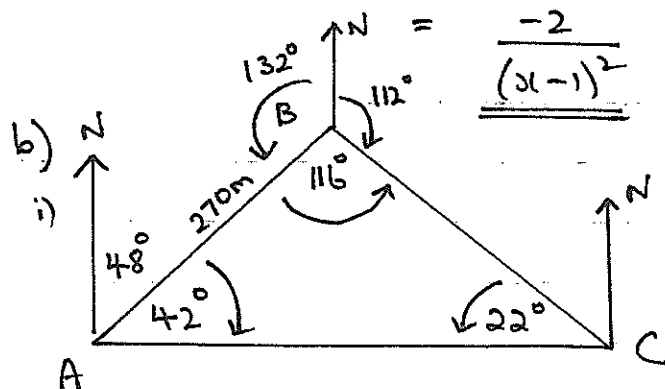
ii) $\frac{d}{dx} (3x^2 - 4)^4 = 4 \cdot 6x (3x^2 - 4)^3$

$$= 24x (3x^2 - 4)^3$$

iii) Let $u = x+1$ $v = x-1$

$u' = 1$ $v' = 1$

$\therefore \frac{d}{dx} \left(\frac{x+1}{x-1} \right) = \frac{1(x-1) - 1(x+1)}{(x-1)^2}$

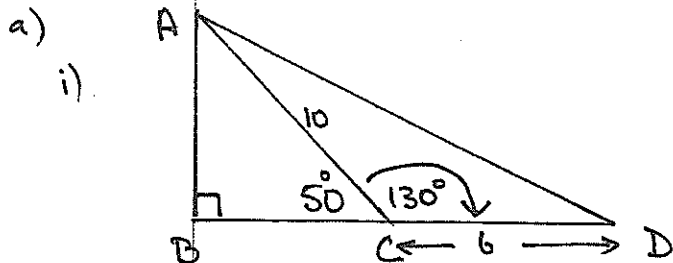


ii) $\frac{AC}{\sin 116^\circ} = \frac{270}{\sin 22^\circ}$

$\therefore AC = \frac{270 \sin 116^\circ}{\sin 22^\circ}$

$AC = 648 \text{ m}$ (nearest m)

Question 13



$$AD^2 = 10^2 + 6^2 - 2 \cdot 10 \cdot 6 \cos 130^\circ$$

$$= 100 + 36 - 120 \cos 130^\circ$$

$$AD = 15 \text{ cm (nearest cm)}$$

ii) Area $\Delta ACD = \frac{1}{2} \cdot 6 \cdot 10 \cdot \sin 130^\circ$

$$= 23 \text{ cm}^2 \text{ (nearest cm}^2\text{)}$$

b) i) sub $k = 2$ into

$$2k^3 - 3k^2 - 4 = 0$$

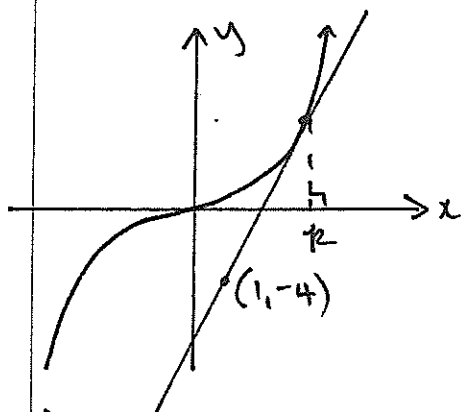
$$\text{LHS} = 16 - 12 - 4$$

$$= 0$$

$$= \text{RHS}$$

$$\therefore k = 2 \text{ is a solution}$$

ii)



$$y = x^3$$

$$\frac{dy}{dx} = 3x^2$$

$$\therefore m_T = 3k^2 \quad \text{where } x = k$$

$$\text{pt } (k, k^3)$$

$$\text{tangent: } y - k^3 = 3k^2(x - k)$$

$$y - k^3 = 3xk^2 - 3k^3$$

$$\underline{y = 3xk^2 - 2k^3}$$

sub $(1, -4)$ into tangent

$$-4 = 3k^2 - 2k^3$$

$$\therefore 2k^3 - 3k^2 - 4 = 0$$

$\therefore$ from part i)

$$\underline{k = 2}$$